

Taut fillings

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Abstract

Sleator, Tarjan, and Thurston asked: given a triangulation σ of the 2-sphere, what is the minimum number of tetrahedra needed to extend σ to a triangulation of the ball? Call this minimum $\text{tetvol}(\sigma)$. Let $X(\sigma)$ be either associated integral 2-cycle, and let $\text{Zvol}(\sigma)$ be the minimum L_1 -norm of an integral 3-chain M with $\partial M = X(\sigma)$. Certainly $\text{Zvol}(\sigma) \leq \text{tetvol}(\sigma)$. We prove that equality holds. More precisely, any taut filling of $X(\sigma)$ is a simplicial chain whose support complex is clean and triangulates B^3 ; the resulting triangulation is freely shellable and flag. The key input is the general fact that taut fillings of integral n -cycles split under disjoint union, connected sum, and more generally under almost disjoint union, where the summands are supported on vertex sets meeting in at most $n + 1$ vertices.

1 Overview

Let $\Delta = \Delta(\Omega)$ be the $|\Omega|-1$ -simplex with vertices Ω , viewed as the abstract simplicial complex consisting of all subsets of Ω . Let $C_n = C_n(\Omega) = C_n(\Delta(\Omega), \mathbf{Z})$ and $Z_n = Z_n(\Omega) = Z_n(\Delta(\Omega), \mathbf{Z})$ be its integral n -chains and n -cycles. A *filling* of an n -cycle $X \in Z_n$ is any $n+1$ -chain $M \in C_{n+1}$ with $\partial M = X$. (Here and throughout we'll assume $n \geq 1$.) Let $\text{Zvol}(X)$ be the minimum L_1 -norm of a filling of X . Call $M \in C_{n+1}$ *taut* if it is an optimal filling of its boundary, i.e., if $|M| = \text{Zvol}(\partial M)$.

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For $X \in C_n$ let $\text{Vert}(X)$ be the set of all the vertices of all the n -simplices to which X assigns non-0 weight. For any taut filling M of X we have $\text{Vert}(M) = \text{Vert}(X)$. (Cf. Proposition 4 below.) This is why we write C_n and Z_n without reference to Ω .

Call an n -cycle $X + Y \in Z_n$ an *almost disjoint union* if

$$|\text{Vert}(X) \cap \text{Vert}(Y)| \leq n + 1.$$

This notion generalizes both disjoint union and connected sum along an n -simplex. Ellison [2] showed the Zvol adds under almost disjoint union: $\text{Zvol}(X + Y) = \text{Zvol}(X) + \text{Zvol}(Y)$. This means that *some* taut filling M of X splits into a sum $M = M_X + M_Y$ of taut fillings of X, Y . Here we amplify Ellison's Corollary 2 to show (Theorem 1) that when $n \geq 2$, *any* taut filling of $X + Y$ splits.

In place of integral chains we can use chains with coefficients in $\mathbf{Q}$ or $\mathbf{R}$. Evidently $\text{Rvol} \leq \text{Qvol}$. But computing Rvol is a linear programming problem with rational coefficients, so in fact $\text{Qvol} = \text{Rvol}$. Ellison used LP duality to prove that Qvol adds under almost disjoint union. Now for $n \geq 2$ we get the stronger result (Corollary 1) that taut $\mathbf{Q}$ -fillings split: Multiplying a taut $\mathbf{Q}$ -filling M of X by a common denominator q yields a taut $\mathbf{Z}$ -filling qM of qX : Split the $\mathbf{Z}$ -filling qM and divide by q to get a splitting of the $\mathbf{Q}$ -filling M .

Now let σ be a simplicial triangulation of S^2 ; let $X(\sigma) \in Z_2(\Delta(\text{Vert}(\sigma)), \mathbf{Z})$ be either one of the 2-cycles that arises from σ by orienting its 2-simplices; and write $\text{Zvol}(\sigma) = \text{Zvol}(X(\sigma))$. Let $\text{tetvol}(\sigma)$ be the minimum number of 3-simplices required to extend σ to a simplicial triangulation τ of B^3 . We will show that

$$\text{Zvol}(\sigma) = \text{tetvol}(\sigma).$$

Certainly $\text{Zvol} \leq \text{tetvol}$, because any extension τ induces a filling of $X(\sigma)$. Theorem 2 states the stronger fact that every taut filling is clean and has support complex a triangulated ball. The proof of Theorem 2 relies on Theorem 1.

We don't know what happens for spheres of dimension 3 or greater.

2 Background

Our interest in Zvol stems from the work of Sleator, Tarjan, and Thurston [5]. They observed that coning from a vertex of maximal degree shows that

a v -vertex triangulation σ of S^2 has $\text{tetvol} \leq 2v - 4 - \max\deg(\sigma)$. If $v \geq 13$ $\max\deg(\sigma) \geq 6$, so $\text{tetvol} \leq 2v - 10$. They produced examples for which $\text{tetvol} = 2v - 10$, provided v is larger than some unspecified bound, and conjectured that such examples should exist for any $v \geq 13$. Their approach was to realize σ as an ideal hyperbolic polyhedron with volume $2vV_0 - O(\log(v))$, where V_0 is the volume of an equilateral ideal tetrahedron, which is maximal. This implies $\text{tetvol} \geq 2v - O(\log(v))$. From there they bootstrapped their way up to showing that $\text{tetvol} = 2v - 10$. They suggested [5, p. 697] that in fact $\text{Qvol} = 2v - 10$, which would immediately imply $\text{tetvol} = 2v - 10$. Mathieu and Thurston [3] produced a class of examples, different from the examples of STT, for which they could show that $\text{Qvol} = 2v - 10$, still under the assumption that v is sufficiently large. In [1] we produced examples for any $v \geq 13$ with $\text{Qvol} = 2v - 10$, confirming the conjecture of Sleator, Tarjan, and Thurston. Theorem 2 here tells us that whenever we have $\text{Qvol} = 2v - 10$, or just $\text{Zvol} = 2v - 10$, any taut filling must arise from a triangulation of the ball.

About Qvol versus Zvol . In [1] we described triangulations of S^2 with $\text{Qvol} < \text{Zvol}$. In all such examples that we know of with $\max\deg \leq 6$, the gap between Qvol and Zvol is < 1 , so that $\lceil \text{Qvol} \rceil = \text{Zvol}$. Since Qvol and Zvol add under connected sum, the gap between them can be arbitrarily large (Ellison [2]), but taking the connected sum pushes $\max\deg$ above 6.

3 Taut fillings

We can think of an n -chain $X \in C_n$ as a multiset of non-cancelling oriented simplices:

$$X = \sum_{t \in X} t.$$

In this sum t denotes an oriented simplex

$$t = [x_0, \dots, x_n] = [x_{\pi(0)}, \dots, x_{\pi(n)}], \quad \pi \text{ an even permutation,}$$

and each unoriented simplex contributes a number of terms corresponding to its multiplicity.

The size of M as a multiset is its L_1 -norm $|M|$. Write $U \subset M$ if U is a sub-multiset of M . This happens just when $|M| = |U| + |M - U|$.

Proposition 1. *If M is taut and $U \subset M$ then U is taut.*

Proof. This is clear, but let's drag it out. If U is not taut, there is a smaller filling V of ∂U . Take $M' = M - U + V$. In terms of multisets, this means that we substitute V for U , and then do any required cancellation of oppositely oriented simplices of $M - U$ and V . We have

$$\partial M' = \partial M - \partial U + \partial V = \partial M$$

and

$$|M'| \leq |M - U| + |V| = |M| - |U| + |V| < |M|,$$

contradiction. $\square$

4 Coning

For $x \in \Omega$, $U \in C_n$ let

$$\text{nbhd}(x, U) = \sum_{t \in U: x \in \text{Vert}(t)} t,$$

$$\deg(x, U) = |\text{nbhd}(x, U)|,$$

$$\text{maxdeg}(U) = \max_x \deg(x, U).$$

The *cone from x to U* is the $(n+1)$ -chain

$$\text{cone}(x, U) = \sum_{t \in U} \text{adj}(x, t),$$

consisting of all the non-trivial oriented $(n+1)$ -simplices $\text{adj}(x, t) = [xx_0 \dots x_n]$ obtained by adjoining x to $t = [x_0 \dots x_n] \in U$. If $x \in \text{Vert}(t)$ then t doesn't contribute to the sum, so $|\text{cone}(x, U)| = |U| - \deg(x, U)$.

If U is closed then $\partial \text{cone}(x, U) = U$, so

Proposition 2. *For any $X \in Z_n$*

$$\text{Zvol}(X) \leq |X| - \text{maxdeg}(X) \quad \square$$

If $U \in Z_n$ and $x \notin \text{Vert}(U)$ then $\deg(x, U) = 0$ and $|\text{cone}(x, U)| = |U|$. In this case we call $\text{cone}(x, U)$ a *complete cone*. By Proposition 2 a non-trivial complete cone is not taut. (This is a long-winded way of saying that instead of coning from $x \notin U$, we could have coned from some $x \in U$.) Thus:

Proposition 3. *If M is taut it contains no non-trivial complete cone.* $\square$

Call $x \in \text{Vert}(M) \setminus \text{Vert}(\partial M)$ an *internal vertex* of M . If x is internal to M then $\text{nbhd}(x, M)$ is a complete cone, so:

Proposition 4. *If M is taut then it has no internal vertices.* $\square$

5 Almost disjoint unions

Recall that if $X, Y \in Z_n$ we call $X + Y$ an almost disjoint union if

$$|\text{Vert}(X) \cap \text{Vert}(Y)| \leq n + 1.$$

The most interesting special case is a connected sum, where $t = [c_0, c_1, \dots, c_n]$ occurs once in X and $-t = [c_1, c_0, \dots, c_n]$ occurs once in Y . For example, if $n = 1$ with X a cycle of length p and Y a cycle of length q the connected sum $X + Y$ is a cycle of length $p + q - 2$. Here Zvol adds, because $\text{Zvol}(X) = p - 2$, $\text{Zvol}(Y) = q - 2$, and

$$\text{Zvol}(X + Y) = (p + q - 2) - 2 = \text{Zvol}(X) + \text{Zvol}(Y).$$

But not every filling of $X + Y$ splits, because a taut filling may contain any 2-simplex with vertices in $\text{Vert}(X + Y)$.

We want to show that fillings do split when $n \geq 2$.

For $A \subset \Omega$, $p \in A$ define

$$\begin{aligned} \pi_{A,p} : \Omega &\rightarrow A \\ \pi_{A,p}(x) &= x \text{ if } x \in A \text{ else } p \end{aligned}$$

and let

$$K_*(A, p) : C_*(\Omega) \rightarrow C_*(A)$$

be the induced chain map. This is a projection of $C_*(\Omega)$ onto $C_*(A)$.

Let A, B be finite vertex sets sharing the vertices $C = A \cap B$, and let $c = |C|$. Observe that $C_c(C)$ and $Z_{c-1}(C)$ are trivial, because $\Delta(C)$ contains no c -simplex, and (up to orientation) only one $c-1$ -simplex, with nothing to cancel its boundary.

Let

$$C_*(A, B) = C_*(A) \oplus C_*(B).$$

For $p, q \in C$ define the mapping

$$g_*(p, q) = K_*(A, p) \oplus K_*(B, q) : C_*(A \cup B) \rightarrow C_*(A, B).$$

Proposition 5. *Let $(X, Y) \in C_n(A, B)$. If $|A \cap B| \leq n$ we can recover (X, Y) from $X + Y$. If $(X, Y) \in Z_n(A, B)$ this holds also when $|A \cap B| = n + 1$.*

Proof. Let $C = A \cap B$. We can assume $|C| \geq 1$. (Add a brand new point to C if necessary.) We claim that for any $p, q \in C$ (not necessarily distinct) we have

$$g_n(p, q)(X + Y) = (X, Y).$$

Indeed,

$$K_n(A, p)(X + Y) = X + K_n(A, p)(Y).$$

The second term belongs to $C_n(C)$, which is trivial when $|C| \leq n$. If $Y \in Z_n(B)$ the second term belongs to $Z_n(C)$, which is trivial when $|C| \leq n + 1$. $\square$

Theorem 1. *If $|A \cap B| \leq n + 1$, for all $(X, Y) \in Z_n(A, B)$ we have*

$$\text{Zvol}(X + Y) = \text{Zvol}(X) + \text{Zvol}(Y).$$

And as long as $n \geq 2$, for any $M \in \text{taut}(X + Y)$ we have $M = M_X + M_Y$ with $M_X \in \text{taut}(X)$, $M_Y \in \text{taut}(Y)$.

Note. Ellison [2] proved the additivity of Zvol . What's new here is the splitting of taut fillings when $n \geq 2$. This result resembles Theorem 6.2 of Pournin and Wang [4] about flip paths. Like theirs, our proof uses a variation on the normalization technique of STT [5, Lemma 7]. It would be nice to fit these results under one roof.

Proof. Again let $CV = A \cap B$. We can assume $|C| = n + 1$, as this is the hardest case. And we might as well go ahead and take $n = 2$, $|C| = 3$, as this case illustrates all the issues.

Take any $M \in \text{taut}(X + Y) \subset C_3(A \cup B)$. Pick distinct points $p, q \in C$, and let

$$(M_X, M_Y) = g_3(p, q)(M).$$

(For now we'll suppress the dependence of M_X, M_Y on p, q .)

Because $g_*(p, q)$ is a chain map we have $\partial_3 M_X = X$, $\partial_3 M_Y = Y$:

$$(\partial_3 M_X, \partial_3 M_Y) = \partial_3(g_3(p, q)(M)) = g_2(p, q)(\partial_3 M) = g_2(p, q)(X + Y) = (X, Y).$$

We want to show that under the map $g_3(p, q)$ any $t \in M$ dies either in M_X or M_Y , because then we'll get additivity of Zvol from

$$\text{Zvol}(X + Y) = |M| \geq |M_X| + |M_Y| \geq \text{Zvol}(X) + \text{Zvol}(Y) \geq \text{Zvol}(X + Y).$$

Let's call a 3-simplex a 'tet' for short. Say that a tet $t \in M$ has type $CCXY$ if $t = \pm[c_1c_2x_1y_1]$ for $c_1, c_2 \in C$, $x_1 \in A \setminus C$, $y_1 \in B \setminus C$. Similarly for types $XXXX$, $CXXX$, $CXXY$, $XXXY$, etc. The first two are pure X cases, meaning that they live in $C_3(A)$; the last two are hybrid cases.

Any $XX..$ tet dies in M_Y ; any $YY..$ tet dies in M_X .

The remaining cases to check are the pure cases $CCCX$, $CCCY$, and the hybrid case $CCXY$. A $CCCX$ tet $t = \pm[c_1c_2c_3x_1]$ dies in M_Y because $K_3(B, q)(t) = [c_1c_2c_3q]$, which vanishes because $q \in C = \{c_1, c_2, c_3\}$ produces a repeated vertex. Likewise a $CCCY$ tet dies in M_X . The more interesting case is $CCXY$: The key is that since $|C| = 3$, $\{c_1, c_2\}$ cannot be disjoint from $\{p, q\}$, so it dies on one side or the other.

So

$$|M| = |M_X(p, q)| + |M_Y(p, q)|,$$

and Zvol adds. Note how we're now emphasizing the possible dependence of M_X, M_Y on p, q . When $n = 1$ the choice of p, q can indeed make a difference: We can get a different pair (M_X, M_Y) if we switch p and q . (Think about the connected sum of two cycle graphs.)

But when $n \geq 2$ we will now show that all tets are pure X or pure Y , so M_X consists of all the pure X tets, and ditto for M_Y . This will make $M = M_X + M_Y$ with $M_X \in \text{taut}(X)$, $M_Y \in \text{taut}(Y)$.

Again our test case is $n = 2$.

The key observation is that, now that we know that every tet dies on one side or the other, we know that no tet can die on both sides, because that would make $|M| > \text{Zvol}(X) + \text{Zvol}(Y)$.

An $XXYY$ tet would die on both sides, so there can be none of these.

Any $pqXY$, $pXXY$, or $qXYY$ would die on both sides, and p, q are arbitrary, so this rules out all $CCXY, CXXY, CXYY$.

The only remaining hybrids are $XXXY$ and $XYYY$. Let's rule out $XXXY$, leaving $XYYY$ to symmetry.

Fix any $y_0 \in B \setminus C$, and suppose there is some $XXXy_0$ tet in M . Let $U \subset M$ consist of all such $XXXy_0$ tets. We claim that $y_0 \notin \text{Vert}(\partial U)$. For let s be any 2-simplex of the form XXy_0 , the only kind of 2-simplex containing y_0 that could belong to ∂U . Any $t \in M$ of which s or $-s$ is a face must be a hybrid tet, and the only possibility is $XXXy_0$, so $t \in U$. Because s has no support in $X + Y = \partial M$, the signed multiplicity of s vanishes in ∂M , hence also in ∂U , because the tets of M with $\pm s$ as a face all belong to U . So $y_0 \notin \text{Vert}(\partial U)$, making U a complete cone on y_0 , contradicting 3. So

there is no such $XXXy_0$ tet, ruling out $XXXY$, and with it $XYYY$.

So there are no hybrid tets, meaning that M splits, as claimed. $\square$

The splitting of taut fillings depends on the assumption $n \geq 2$. Let's consider what goes wrong when $n = 1$. Here the hybrid types are CXY , XXY , XYX . Neither pXY nor qXY dies on both sides, so we can't rule out CXY . Failing that, the XXy_0 tets needn't form a complete cone, because $[x_0x_1y_0]$ can continue across $[x_1y_0]$ to $-[px_1y_0]$, so we can't rule out XXY either.

The foregoing proof works equally well over $\mathbf{Q}$, providing we permit ourselves to work with fractional multisets. Alternatively, we can clear denominators as in the introduction above. Either way, we have:

Corollary 1. *$\mathbf{Q}\text{vol}$ adds under almost disjoint union, and for $n \geq 2$ taut $\mathbf{Q}$ -fillings split.* $\square$

6 Triangulations

We turn now to filling simplicial triangulations of S^2 . We first fix the combinatorial language.

An n -simplex is a set of size $n + 1$. Its faces are its subsets, regarded as simplices of dimensions $-1, 0, \dots, n$, where the empty simplex has dimension -1 . A *simplicial complex* K is a finite collection of simplices closed under taking faces: if $t \subset s \in K$, then $t \in K$.

For a simplex $s \in K$, define the *link*

$$\text{link}(s, K) = \{t : t \cap s = \emptyset, s \cup t \in K\}.$$

This is again a simplicial complex. Except when $s = \emptyset$, it is not literally a subcomplex of K : it records the faces complementary to s in simplices containing s .

We will call an n -dimensional simplicial complex *clean* if it is a normal pseudomanifold with boundary, in the following concrete sense. First, it is pure: every simplex belongs to some n -simplex. Second, every $(n - 1)$ -simplex lies in either one or two n -simplices; those lying in only one form the boundary. Third, for every simplex s of dimension $0, \dots, n - 2$, the link $\text{link}(s, K)$ is connected. Thus in dimension 3 we are requiring the usual triangle-count condition together with connected edge links and connected vertex links. We do not impose connectedness of K itself as a separate

convention; in the applications below the complexes under discussion are connected.

The boundary of a clean complex is clean and closed, and links in a clean complex are clean of the corresponding dimension.

If N is a manifold, a simplicial complex K is a *simplicial triangulation* of N if its geometric carrier is homeomorphic to N . When N is oriented, a choice of orientation of the triangulation gives an integral cycle $X(K)$; in the case $N = S^2$ the two choices differ by sign, and nothing below depends on which one is chosen.

This notion is the ordinary simplicial one. We do not allow two distinct simplices with the same vertex set, nor do we allow self-identifications of a simplex along its own faces.

We continue to regard an n -chain $M \in C_n$ as a multiset of oriented n -simplices. We call M *simplicial* if no unoriented simplex occurs with multiplicity greater than one. If M is simplicial, let $K(M)$ denote the pure simplicial complex generated by the unoriented simplices occurring in M . A filling M is a chain, not a complex; we will nevertheless call M *clean* if M is simplicial and $K(M)$ is a clean complex.

7 Filling a triangulation of the 2-sphere

Theorem 2. *Let σ be a simplicial triangulation of S^2 , and let M be a taut filling of $X(\sigma)$. Then M is clean, and $K(M)$ is a simplicial triangulation of B^3 .*

Proof. Call (σ, M) a *filling pair* if σ is a simplicial triangulation of S^2 and M is a taut filling of $X(\sigma)$. Call such a pair *good* if M is clean and $K(M)$ is a simplicial triangulation of B^3 . We prove that every filling pair is good.

Suppose not, and choose a bad filling pair (σ, M) for which $|M|$ is minimal. Let v, e, f be the numbers of vertices, edges, and faces of σ ; then $e = 3v - 6$ and $f = 2v - 4$. The tetrahedron boundary is the base case, so $v > 4$. Since taut fillings split under connected sum, we may assume that σ is prime. In particular, the only consequence we shall use is that σ has no vertex of degree 3.

A tet $t \in M$ is *eligible* if two of its oriented faces occur as faces of σ with the boundary orientation. It cannot have more than two such faces, because

that would give a degree-3 vertex of σ . By Proposition 2,

$$|M| = \text{Zvol}(\sigma) \leq f - \maxdeg(\sigma).$$

For each support tet u of M , let $A(u)$ be the set of faces of σ which occur as properly oriented faces of u . These sets cover the faces of σ , since each boundary face has nonzero coefficient in ∂M with the sign specified by $X(\sigma)$. Moreover $|A(u)| \leq 2$. Reveal the support tets one at a time, and count how many previously uncovered boundary faces each reveals. If r tets reveal two new boundary faces, then

$$f \leq |M| + r.$$

Hence $r \geq \maxdeg(\sigma)$. These r tets are eligible, and the pairs of new boundary faces assigned to them are disjoint. We shall use only the existence of two such eligible tets, and the fact that for any fixed boundary face there is an eligible tet not containing it.

Choose an eligible tet t with boundary faces

$$s_1 = [a, b, c], \quad s_2 = [b, a, d].$$

Removing t replaces these two boundary faces by

$$[c, d, b], \quad [d, c, a],$$

which is the boundary flip replacing the edge ab by the edge cd . Let the resulting boundary complex be denoted by σ_t . There are two possibilities.

In the first case, cd was not an edge of σ , and σ_t is again a simplicial triangulation of S^2 . Then $M - t$ is a taut filling of σ_t , so by minimality $(\sigma_t, M - t)$ is good. Thus $M - t$ is clean and $K(M - t)$ triangulates B^3 . Gluing t back along the two exposed boundary faces gives $K(M)$, provided t was not already present in $M - t$. If t were already present, then M would contain the same unoriented tet twice. Choose another eligible tet u whose two boundary faces are disjoint from those of t . Removing u leaves a smaller filling in which the doubled tet t is still present, so the smaller pair is still bad, contradicting the minimality of (σ, M) . Hence no such doubling occurs, and in this case M is good.

In the second case, cd was already an edge of σ . Then σ_t is an almost disjoint union of two triangulated spheres, say σ_1 and σ_2 , joined along the edge cd . The filling $M - t$ need not be clean as one complex: the link of

the common edge is disconnected. But by Theorem 1, the taut filling $M - t$ splits as

$$M - t = M_1 + M_2,$$

where M_i is a taut filling of σ_i . By minimality, each (σ_i, M_i) is good. Thus $K(M_1)$ and $K(M_2)$ are clean triangulated balls. Gluing these two balls together with the bridging tet t gives $K(M)$, again unless t was already present in $M - t$. The same second eligible tet argument rules this obstruction out. Therefore M is good in this case as well.

It remains only to spell out why this gluing preserves cleanness. Away from the two faces along which t is attached, the links and face incidences are unchanged from the smaller clean ball or balls. Along the gluing faces, each boundary triangle becomes incident to exactly two tets. The new local edge and vertex links are obtained from connected intervals or disks by attaching one simplex along a nonempty connected part, so they remain connected. The only possible failures would be a repeated tet, a triangle used by more than two tets, or a disconnected edge or vertex link already visible after removing a disjoint eligible tet. Such a persistent failure would make the smaller pair bad, contradicting the chosen minimality. Hence the reassembled complex is clean and triangulates B^3 .

This contradiction proves the theorem. $\square$

Thanks to this theorem, we may take the liberty of saying that a taut filling of σ is a simplicial triangulation τ of B^3 .

8 Shelling

We next show that a taut filling may be shelled from any chosen initial tet. A *shelling* of a simplicial triangulation τ of B^3 is an ordering $(t_1, \dots, t_{|\tau|})$ of its tets such that, for every k , the subcomplex generated by $t_1, \dots, t_k$ is again a simplicial triangulation of B^3 . For $k > 1$, the new tet t_k may meet the previous complex in one, two, or three faces. In our situation the three-face case is excluded by the absence of internal vertices. We call τ *freely shellable* if every tet of τ can be chosen as the first tet of some shelling.

Theorem 3. *If σ is a simplicial triangulation of S^2 , any taut filling τ of σ is a freely shellable simplicial triangulation of B^3 .*

Proof. This follows by turning the induction in Theorem 2 into a recursive “shucking” procedure. Fix the tet t_1 which is to be first in the final shelling;

equivalently, we will remove all other tets first and remove t_1 last.

If τ consists only of t_1 , there is nothing to prove. If a degree-3 vertex belongs to a tet $t \neq t_1$, remove that tet and shuck the remaining filling, still ending with t_1 . If the only available degree-3 vertex belongs to t_1 , let t_2 be the tet of $\tau - t_1$ to which t_1 is attached; shuck $\tau - t_1$ ending with t_2 , and then put t_1 first when reading the removals backwards.

If there is no degree-3 vertex, the eligible-tet count supplies at least $\maxdeg(\sigma) \geq 4$ disjoint eligible tets. Choose one sharing no face with t_1 . If removing it leaves one sphere boundary, recurse on the smaller filling. If removing it splits the boundary into two conjoined spheres, split the filling as in Theorem 1, shuck the side not containing t_1 first, then remove the bridging tet, and finally shuck the side containing t_1 . Reading this removal order backwards gives a shelling beginning with t_1 . $\square$

9 Flag complex

A simplicial complex is a *flag complex* if every clique in its 1-skeleton occurs as a simplex. A triangulation of S^2 is flag exactly when it is prime and is not the boundary of a tetrahedron.

We can decompose any triangulated 2-sphere as a connected sum of prime components. Taut fillings split along this connected-sum decomposition, so it is enough to understand the prime pieces; nevertheless the conclusion below is stated for arbitrary σ .

Theorem 4. *If σ is a simplicial triangulation of S^2 , any taut filling τ of σ is a flag complex.*

Proof. We must rule out three taboo configurations in τ :

1. the three edges of a triangle $\{A, B, C\}$ without the face ABC ;
2. the six edges of a K_4 on $\{A, B, C, D\}$ without the tet $ABCD$;
3. the ten edges of a K_5 on $\{A, B, C, D, E\}$.

If the first two configurations are ruled out, the third is ruled out as well: a K_5 would force all five of its K_4 subcomplexes to occur as tets, forming the boundary of a 4-simplex, which cannot occur inside a triangulated 3-ball.

Suppose there is a minimal counterexample. If σ has a vertex of degree 3, then σ is a connected sum with one summand the boundary of a tetrahedron.

Removing the corresponding tet from τ gives a smaller counterexample, a contradiction. Thus we may assume that σ has no vertex of degree 3.

Then the eligible-tet count gives at least $\maxdeg(\sigma)$ disjoint eligible tets. Removing one performs a boundary edge flip. The new boundary either remains a triangulated sphere or becomes an almost disjoint union of two spheres joined along an edge. In the first case, preserving the taboo configuration gives an immediate smaller counterexample. In the second case, a taboo K_3 or K_4 in the almost disjoint union lies entirely on one side or the other, again giving a smaller counterexample after splitting the taut filling.

For a missing triangle, only one edge disappears when we remove an eligible tet: the boundary edge being flipped. There are at least four eligible flips, and only the three edges of the taboo triangle need to be avoided. Thus some flip preserves the missing triangle, contradicting minimality.

For a missing tet on vertices A, B, C, D , there are six edges to avoid. Since missing triangles have already been ruled out, all four faces ABC, ABD, ACD, BCD occur. At most two of these faces can lie in the boundary σ , because otherwise σ would have a degree-3 vertex. Treating the octahedral boundary as the small exceptional case, we may assume $\maxdeg(\sigma) \geq 5$. Hence there are at least five disjoint eligible flips. If five of the six K_4 edges are not simultaneously boundary edges, one of these flips avoids the taboo K_4 . In the remaining case, say the only interior edge is AD , so the boundary contains the faces ABC and BCD . At most two of the disjoint eligible tets can use one of these two faces, leaving another eligible flip which preserves the K_4 . Again we get a smaller counterexample, contradiction. $\square$

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